Please stick the barcode label here.

5. Figure 5.1 shows a smooth sloping track *ABC* firmly fixed in a vertical plane with its horizontal part *BC* resting on a bench surface. You are given a toy skier, a metre rule and a long rough paper strip with adhesive tape on the bottom surface.

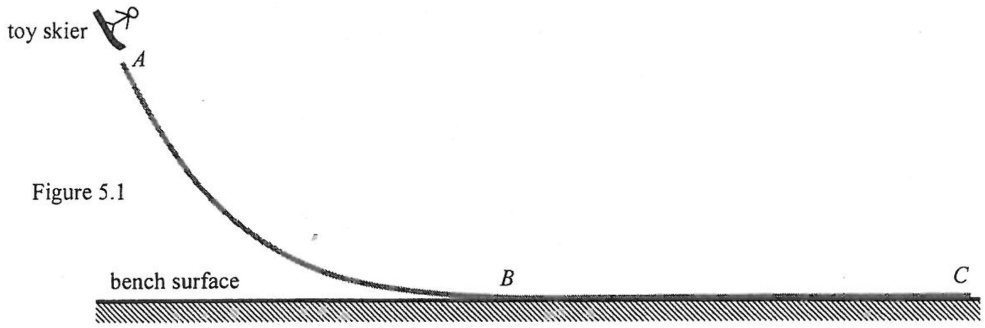

Figure 5.1

Using the apparatus provided, describe an experiment to study how the stopping distance of the toy skier depends on its height of release. Your description should include the physical quantities to be measured and the result expected. (5 marks)

8. In Figure 8.1, $A'B'$ represents the image of an object AB formed by a lens L (not shown) where XY is the principal axis of the lens.

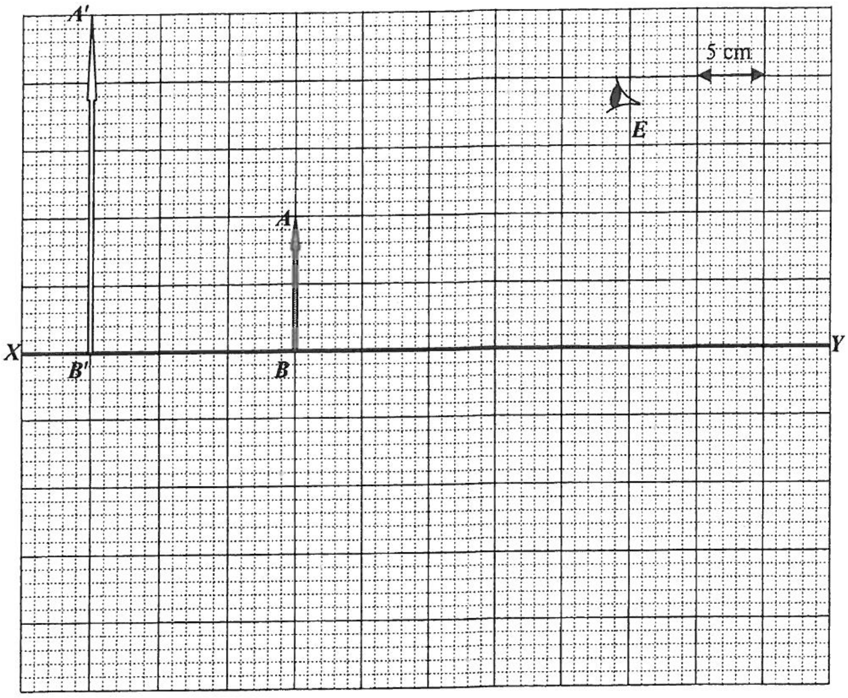

Figure 8.1

(a) (i) Is the image real or virtual ? (1 mark)

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(ii) What kind of lens is used ? Explain your answer. (2 marks)

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(b) (i) Locate the optical centre O of lens L and draw on Figure 8.1 the position of lens L. (1 mark)

(ii) By drawing an additional light ray, mark the principal focus F of the lens and find its focal length. The horizontal scale is 1 cm to 5 cm. (2 marks)

Focal length = ...

(c) Draw a light ray to show how the eye E shown can see the image of head A through lens L. (2 marks)

(d) State an application of lens L in the situation as shown above. (1 mark)

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(b) Describe what would be observed if the experiment in part (a) is repeated with

(i) a low voltage a.c. supply; (1 mark)

(ii) a low voltage a.c. supply and an aluminium ring with a slit cut through it as shown []. (1 mark)

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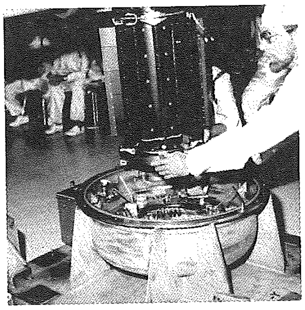
10. *Voyager I* is a space probe designed by NASA to operate for over ten years in space. It was equipped with a radioisotope thermoelectric generator (RTG) which can convert the energy released from the decay of a radioactive source into electrical power. *Voyager I* operates with a plutonium-238 radioactive source that undergoes α-decay.

(a) The plutonium-238 source is sealed inside a thin metallic casing of the RTG. The photo shows a NASA staff handling the RTG with his bare hands. Explain why it is fine for it to be handled by the staff in this way. (1 mark)

Figure 10.1



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When *Voyager I* was launched, the number of plutonium-238 atoms in the source was 3.2×10^{25}.
Given: half-life of plutonium-238 = 87.74 years.
Take 1 year = 3.16×10^{7} s.

(b) *(i) Find the activity, in Bq, of the plutonium source at the time of launch. (3 marks)

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(b) (ii) When a plutonium-238 atom decays, it releases 5.5 MeV of energy. Estimate the power, in kW, delivered by the source at the time of launch. (2 marks)

*(iii) The RTG of *Voyager I* is still in operation as *Voyager I* just left the solar system in September 2013 after it was launched 36 years ago. Estimate the corresponding power delivered by the plutonium source, expressed in **percentage** of the power delivered at the time of launch. (2 marks)

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(b) At a certain place the Earth's magnetic field runs along the S-N direction such that the field lines make an angle θ with the horizontal as shown in Figure 9.2(a).

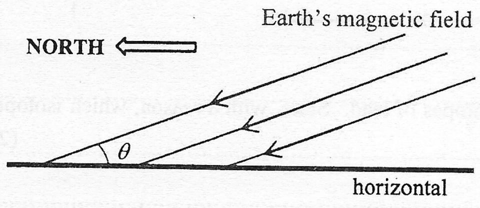

Figure 9.2(a)

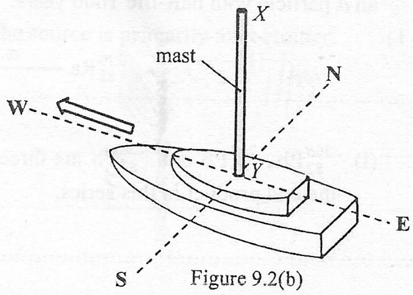

Figure 9.2(b)

A ship with a vertical aluminium mast sails at sea along a straight course to the west as shown in Figure 9.2(b). As a result, an e.m.f. is induced across the mast XY.

(i) Explain why it is **only the horizontal component** of the Earth's magnetic field that is cut by the mast which gives rise to this induced e.m.f. (1 mark)

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(ii) Given: length of mast XY = 20 m
speed of the ship = 6 m s^{-1}
Earth's magnetic field = 50 μT
θ = 30°

Referring to (a)(iii), calculate the e.m.f. induced across XY and state whether the distribution of free electrons along the mast is more at end X, more at end Y or uniform along XY. (3 marks)

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(iii) Suppose X and Y are connected by a cable running side-by-side with the mast so that they form a complete circuit. Explain whether there will be a current passing through it. (2 marks)

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10. (a) Part of the decay series of radium-226 (Ra-226) is shown below. Ra-226 decays to radon (Rn) by emitting an α particle with half-life 1600 years. The end product in the series is lead (Pb), which is stable.

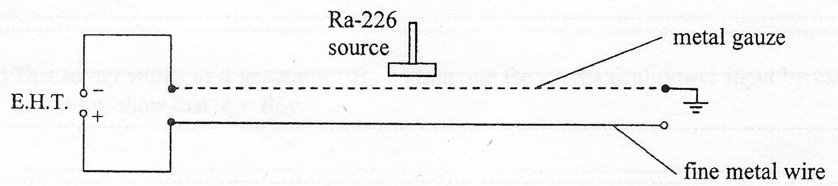
$$^{226}_{88}\text{Ra} \xrightarrow{\alpha} \text{Rn} \longrightarrow \cdots \longrightarrow \text{Pb}$$

(i) $^{206}_{82}\text{Pb}$, $^{207}_{82}\text{Pb}$ and $^{208}_{82}\text{Pb}$ are three stable isotopes of lead. State, with a reason, which isotope can be the end product in this series. (2 marks)

*(ii) In a certain laboratory, a Ra-226 source has been used for 50 years. Estimate the percentage of undecayed Ra-226 left after this period. (2 marks)

(b) Spark counters can show the ionizing power of radiations. Figure 10.1 indicates the main features of a type of spark counters in school laboratories.

Figure 10.1



A spark counter consists of a fine metal wire mounted a few mm beneath an earthed metal gauze. The wire is connected to the positive terminal of an E.H.T. (Extra High Tension) supply so that a very intense electric field is set up between the wire and the metal gauze. When a Ra-226 source is brought near the gauze, sparks giving out flashes of light and crackling sound are produced at irregular intervals.

(i) Explain why the sparks occur at **irregular** intervals. (1 mark)

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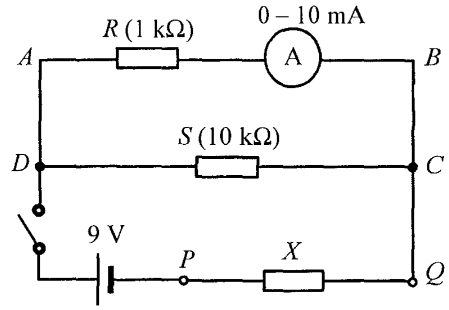

Figure 8.1

Figure 8.1 shows a circuit for measuring the resistance of resistor X connected across P and Q. The resistance of resistor S is 10 kΩ. The internal resistance of the 9 V cell and that of the ammeter are negligible.

(a) When the switch is closed, the ammeter reads 8.5 mA.

(i) What is the p.d. between A and B ? (2 marks)

(ii) Find the current passing through resistor S. (2 marks)

(iii) Indicate on Figure 8.1 the direction of current in each of the three branches via C. (2 marks)

(iv) Deduce the p.d. across resistor X. Hence, find the resistance of X. (3 marks)

(b) State the purpose of connecting resistor R in series with the ammeter. (1 mark)

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9. Potassium-40 ($^{40}_{19}K$) is a natural radioisotope of potassium.

(a) (i) What kind of decay does $^{40}_{19}K$ undergo if it decays to $^{40}_{20}Ca$? (1 mark)

(ii) As banana is rich in potassium, a student claims that the radiation emitted by $^{40}_{19}K$ after eating a few bananas can be detected outside the human body. Explain whether this claim is justified. (1 mark)

*(b) A banana typically contains 0.45 g potassium in which 0.012% by mass is $^{40}_{19}K$ while the rest is $^{39}_{19}K$.

Given: half-life of $^{40}_{19}K = 1.25 \times 10^9$ years
1 year = 3.16×10^7 seconds
molar mass of $^{40}_{19}K = 40.0$ g

(i) Estimate the number of moles of $^{40}_{19}K$ in a banana. (1 mark)

(ii) Deduce the activity, in Bq, of a banana. (2 marks)

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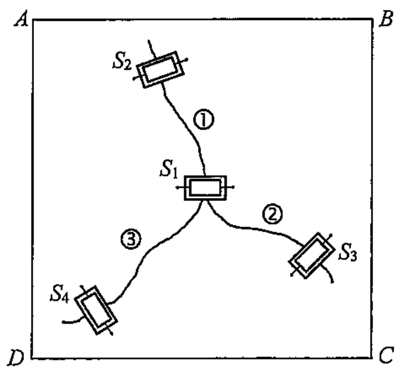
(i) Current I_1 flows through branch ABC. Write an equation relating V_{in}, I_1, R_1 and R_2. (1 mark)

(ii) R_4 in the circuit is the sensing device described in the passage, with unstretched resistance 500 Ω. Find the corresponding percentage change in V_{out}/V_{in} when R_4 increases 0.2% (i.e. becomes 501 Ω).

Given: $\frac{V_{out}}{V_{in}} = \frac{R_4}{R_3 + R_4} - \frac{R_2}{R_1 + R_2}$; $R_1 = R_2 = R_3 = 470\ \Omega$. (2 marks)

(iii) Sensing devices S_1, S_2, S_3 and S_4 connected to a suitable circuit are attached to ceiling $ABCD$ as shown in Figure 11.3 in order to monitor the growth of the cracks ①, ② and ③. After a few weeks, the extent of deformation detected is in the order $S_1 > S_2 > S_3 > S_4$.

Figure 11.3



(I) Deduce which crack (①, ② or ③) has grown the most. (1 mark)

(II) Deduce the most probable stretching direction (AB, BC or AC) along the ceiling. (1 mark)

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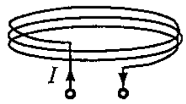
12. Figure 12.1 shows a coil of several turns.

Figure 12.1



(a) Sketch the magnetic field pattern produced by the coil when it carries a current I as shown. (2 marks)

(b) As shown in Figure 12.2(a), this coil is now connected to an a.c. source to serve as a transmitter coil (T). A receiver coil (R) connected to a CRO is placed directly above T. The time variation of the voltage of coil R is displayed on the CRO (Figure 12.2(b)).

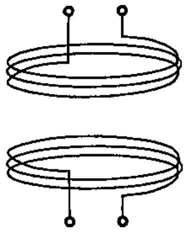


Figure 12.2(a)

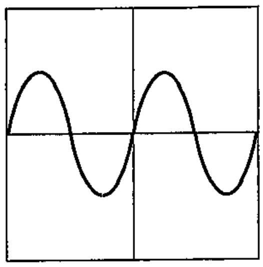


Figure 12.2(b)

(i) Explain how the voltage of the receiver coil R is produced. (2 marks)

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*(ii) On Figure 12.2(b), sketch the CRO display when the frequency of the a.c. source is doubled. (2 marks)

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(c) A mobile phone with a receiver coil inside can be charged wirelessly by placing it on a charging pad equipped with a built-in transmitter coil, which is connected to a power source.

Explain why wirelessly charging of phones need phone cases with non-metallic back covers. (2 marks)

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13. Radon (Rn-222) is a radioactive gas with no colour, no smell and no taste. It is released from bedrock material underground or from concrete composed of granite. Rn-222 decays with a half-life of 3.82 days by giving out α particles.

(a) *(i) Find the decay constant (unit: s^{-1}) of Rn-222. (2 marks)

(ii) Inside buildings, the average activity due to radon in 1 m^3 of air is about 48 Bq. Estimate the average number of radon atoms in 1 m^3 of air inside buildings. (2 marks)

(iii) Why is indoor radon considered harmful to human although α particles have limited penetrating power ? (2 marks)

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(b) Transmission cables at 400 kV are used for long-distance electric power transmission to a city.

(i) Estimate the current in the cables. (2 marks)

(ii) Estimate the power loss (in kW) in the transmission cables. The resistance of the cables is $2.90 \times 10^{-8}\ \Omega$ per metre and the total length of the cables is 160 km. (2 marks)

*(iii) Explain why a.c. is preferred for electric power transmission. (2 marks)

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12. Carbon-14 ($^{14}_{6}C$) nuclide decays into stable nitrogen-14 ($^{14}_{7}N$) nuclide with a half-life of 5730 years. Assume that the concentration of carbon-14 in the atmosphere remains unchanged over time.

(a) (i) Write an equation to represent the decay of carbon-14. (1 mark)

*(ii) Find the decay constant (in yr^{-1}) of carbon-14. (2 marks)

*(iii) 1 g of an ancient wood sample has a carbon-14 activity that gives a count rate of 6.0 counts per minute. Estimate the age of this sample in years. Given that the carbon-14 activity in 1 g of fresh wood gives a count rate of 13.6 counts per minute. (2 marks)

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(b) A more advanced method involves directly measuring the number of carbon-14 atoms relative to that of the carbon-12 atoms. Carbon atoms from the wood sample are ionized to form a beam which is directed into a magnetic field as shown in Figure 12.1. Each of the positive carbon-14 and carbon-12 ions possesses one electronic charge in magnitude.

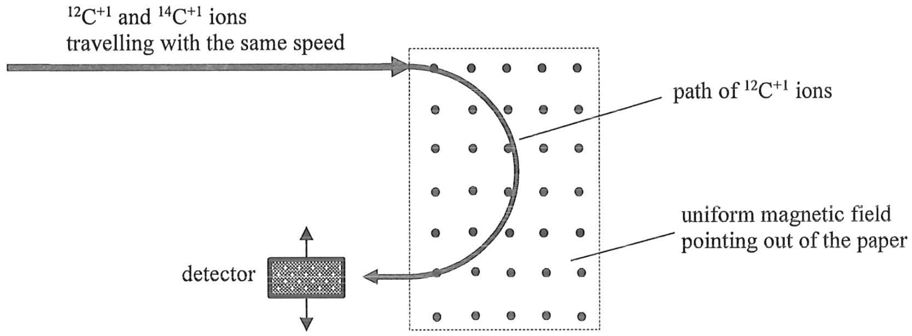

Figure 12.1

(i) Both carbon-14 and carbon-12 are isotopes of carbon. What is meant by 'isotopes' ? (1 mark)

*(ii) Sketch the path of $^{14}C^{+1}$ ions on Figure 12.1. (1 mark)

(iii) Explain why the speed of the ions remains unchanged after passing through the magnetic field. (2 marks)

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